SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DSA0105 - Object Oriented programming with C++

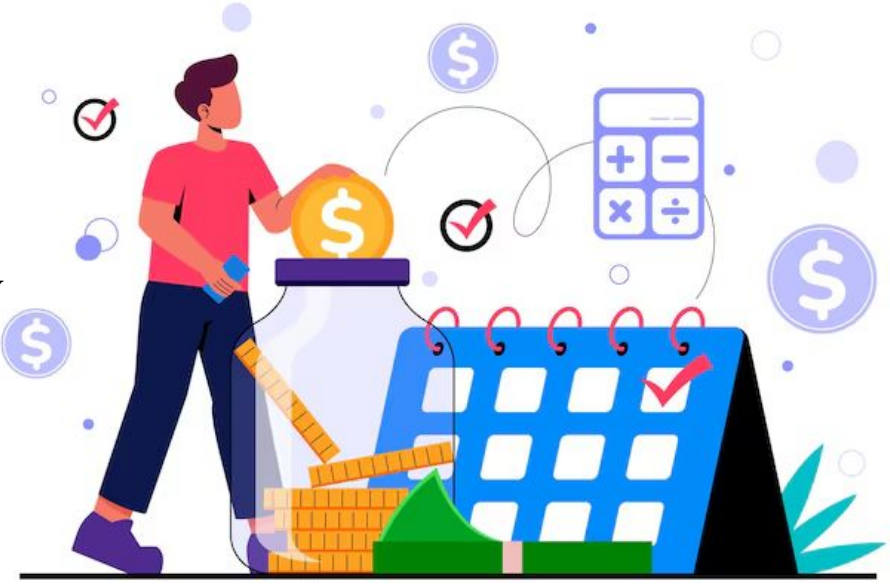
Personal Expense Tracking and Smart Budget Planning Application Using C++

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PROBLEM STATEMENT

- People often find it difficult to track their daily income and expenses accurately.
- Manual calculations make it harder to understand spending habits and manage monthly budgets.
- Users may overspend because they do not receive timely alerts when their expenses approach budget limits.
- A simple application is needed to record transactions, monitor budgets, and provide clear financial summaries in one place.



Existing System vs Proposed System

Existing System

- Expenses are tracked manually or using basic spreadsheets.
- Budget calculations and spending analysis require manual effort.
- Overspending may go unnoticed without timely alerts.

Proposed System

- Records and categorizes income and expenses in one application.
- Automatically calculates budgets, balances, and spending totals.
- Provides overspending alerts and visual financial reports.

INTRODUCTION

- Managing personal finances is important for controlling expenses and maintaining savings.
- This application allows users to record their income and daily expenses manually.
- It helps users categorize spending, set monthly budgets, and receive overspending alerts.
- The application provides financial summaries and charts through a responsive interface accessible on laptops and mobile phones.



Objectives

- To develop a simple application for managing personal income and expenses.
- To categorize transactions and simplify expense tracking.
- To monitor budgets and generate automatic overspending alerts.
- To provide financial summaries and charts for better financial planning.



Technologies and Techniques Used

Frontend

- HTML5, CSS3
- JavaScript
- Chart.js

Backend

- C++17
- cpp-httplib (HTTP server)
- nlohmann/json (JSON processing)

Database

- SQLite

Operating System

- Windows 10 / Windows 11

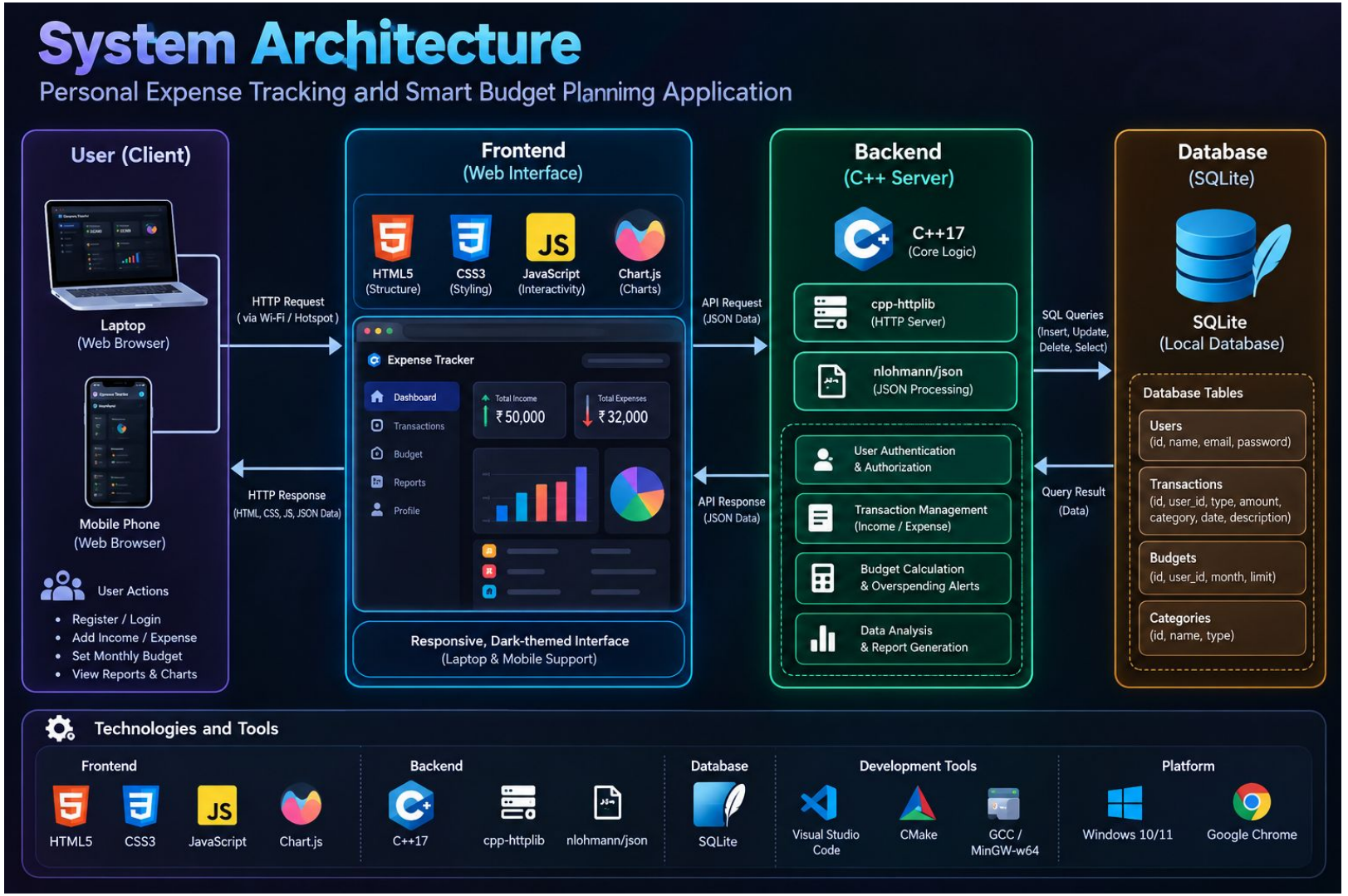
Web Browser

- Google Chrome

Techniques Used

Transaction management and expense categorization
Search and filtering
Budget calculation and overspending detection
Financial summaries and chart-based reporting

SYSTEM ARCHITECTURE



Module 1 – User Accounts and Transaction Management

- Allow users to register and log in to the application.
- Record income and expenses with amount, date, and description.
- Allow users to view, edit, and delete transactions.
- Store transaction details securely in the SQLite database.

Module 2 – Expense Categorization and Search

- Organize expenses into categories such as food, transport, education, and shopping.
- Search transactions using keywords and amounts.
- Filter records by date, category, and transaction type.
- Display organized transaction history for easy tracking.

Module 3 – Budget Planning and Overspending Alerts

- Allow users to set monthly and category-wise budgets.
- Calculate total spending against the defined budget.
- Display the remaining budget and percentage spent.
- Generate alerts when spending approaches or exceeds the budget limit.

Module 4 - Financial Summaries and Spending Reports

- Calculate total income, total expenses, and remaining balance.
- Display category-wise spending using charts.
- Generate monthly financial summaries and comparisons.
- Help users understand spending habits and manage their finances better.

CONCLUSION

- The application provides a simple way to record and manage personal income and expenses.
- It helps users plan budgets, monitor spending, and receive overspending alerts.
- Financial summaries and charts make spending patterns easier to understand.
- The C++ backend, SQLite database, and responsive dark interface provide a practical solution accessible on laptops and mobile phones.

THANK YOU